

Module 2: Network + Network Security , Maintenance and Troubleshooting

Day 1 Introduction to Network Protocols

Task 1:

328	7.087722	192.168.1.5	192.168.1.1	DNS	75	Standard query 0xf6fa HTTPS www.spotify.com
329	7.087816	192.168.1.5	192.168.1.1	DNS	75	Standard query 0x3083 A www.spotify.com

DNS Packets

343	7.177121	192.168.1.5	151.101.107.42	TLSv1...	583	Client Hello (SNI=www.spotify.com)
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HTTPS Packets

Task 2:

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aashanapatel@192 ~ % nslookup www.flipkart.com
Server:          192.168.1.1
Address:         192.168.1.1#53

Non-authoritative answer:
www.flipkart.com canonical name = flipkart.com.
Name:   flipkart.com
Address: 163.53.76.86
```

Task 3:

	HTTP	HTTPS	Example
Encryption	Data sent in plaintext	Data encrypted via TLS/SSL	Online banking apps (e.g., an HDFC or SBI banking app) rely
Port	Port 80	Port 443	

	HTTP	HTTPS	Example
Security	Vulnerable to eavesdropping/tampering	Verifies server identity via certificates and protects data in transit	entirely on HTTPS — when you enter your login credentials or transaction PIN, TLS encryption prevents anyone intercepting your Wi-Fi traffic from reading that data.

Task 4:

2716	192.168.1.5	192.168.1.1	DNS	77	Standard query	0x9b87	HTTPS	www.pinterest.com
2813	192.168.1.5	192.168.1.1	DNS	77	Standard query	0x4502	A	www.pinterest.com
7828	192.168.1.1	192.168.1.5	DNS	282	Standard query response	0x9b87	HTTPS	www.pinterest.com CNAME
8455	192.168.1.1	192.168.1.5	DNS	240	Standard query response	0x4502	A	www.pinterest.com CNAME

Task 5:

Real-world protocol scenario — DHCP example

Scenario: You walk into a café and open Instagram on your phone. Your phone joins the café's Wi-Fi, and before it can load anything, it needs an IP address. It sends a **DHCP Discover** broadcast; the café's router (acting as a DHCP server) responds with a **DHCP Offer** containing an available IP address, subnet mask, gateway, and DNS server info. Your phone sends a **DHCP Request** to accept that offer, and the router confirms with a **DHCP Ack**. Only after this handshake does your phone have a usable IP address and can start resolving Instagram's domain via DNS and loading your feed over HTTPS.